

Grounding Image Captions: Reducing Hallucination in Vision-Language Models

Final Report

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Abstract

Image captioning models often mention objects that are not present in the image. We treat this as a caption *selection* problem rather than a generation problem. For each image we generate six candidate captions using BLIP, then pick the one that scores highest under a simple hybrid function: CLIP visual alignment minus an object-count penalty and a length penalty. No retraining is required. On 5,000 COCO val2017 images, our method reduces CHAIRs from 4.08% to 0.74%, an 82% relative reduction. An ablation isolates the object penalty as the dominant component and shows that CLIP alignment alone actually *increases* hallucination.

1 Introduction

This milestone scales our Milestone 2 pipeline from 500 to 5,000 COCO val2017 images, adds an ablation study over the hybrid scoring weights, and refines the CHAIR-style evaluation. The method itself is unchanged: BLIP generates one greedy caption and five nucleus-sampled captions per image, and a hybrid scoring function picks the best one. The contribution of Milestone 3 is to show that the Milestone 2 result holds at scale, and to explain *why* the hybrid works through component-level ablation.

2 Method

For each image, BLIP generates a candidate pool of six captions: one greedy caption (sample=False, num_beams=1) and five nucleus-sampled captions (top_p=0.9). Each candidate is scored as

$$Score = \alpha \cdot CLIP(img, cap) - \beta \cdot ObjectPenalty(cap) - \gamma \cdot LengthPenalty(cap)$$

with $\alpha = 1.0$, $\beta = 0.3$, $\gamma = 0.2$ as the default weights. These weights were tuned empirically in Milestone 2 to balance visual alignment with grounding constraints. The CLIP score is the cosine similarity between the openai/clip-vit-base-patch32 image and text embeddings, normalised to $[0, 1]$. The object penalty is the count of unique COCO-category mentions divided by 10. The length penalty is $\max(0, |words| - 10)/20$. The penalties are intentionally lightweight heuristics designed to discourage overly descriptive captions that are more likely to hallucinate unsupported objects. The caption with the highest score becomes the final output for that image.

All experiments use random seed 42 across random, numpy, torch, and CUDA. We use BLIP’s pretrained model_base_caption_capfilt_large weights without modification. The method is entirely training-free.

3 Experiments and Results

3.1 Setup

We evaluate on the first 5,000 images of COCO val2017 (sorted by filename), the same selection rule as Milestone 2 for direct comparability. Ground truth is constructed by combining segmentation

annotations from instances_val2017.json with objects extracted from captions_val2017.json, mapped to COCO categories using the official synonym list of 403 entries from the CHAIR repository. Words are singularised with inflect before matching.

We compare three caption-selection strategies. *Baseline* keeps the greedy BLIP caption. *CLIP-only* picks the candidate with the highest CLIP score. *Hybrid* picks the candidate with the highest score under our weighted function above.

3.2 Main Results

Table 1 reports CHAIRs (sentence-level) and CHAIRi (instance-level) for all three methods. The hybrid method reduces CHAIRs from 4.08% to 0.74%, an 82% relative reduction. On absolute counts, the number of hallucinated captions falls from 204 to 37 out of 5,000.

Table 1: Main results on 5,000 COCO val2017 images. Lower is better.

Method	CHAIRs	CHAIRi	Avg. length	Avg. objects
Baseline (greedy)	4.08%	2.63%	9.69	1.59
CLIP-only selection	5.10%	3.48%	10.24	1.51
Hybrid (ours)	0.74%	0.74%	9.25	1.02

CLIP-only selection performs *worse* than the baseline at 5.10% CHAIRs. This was unexpected at first, but the reason is straightforward: when CLIP is the only signal, it tends to prefer the most descriptive-sounding caption in the pool, and longer captions name more objects, giving more chances to hallucinate. This finding directly justifies the two penalty terms in the hybrid.

We also note that the hybrid reduces the average number of object mentions per caption from 1.59 to 1.02, a 36% drop. However, the 82% reduction in CHAIRs is much larger than this 36% drop in object count. This disproportionate improvement indicates the hybrid is not simply choosing shorter captions; it is actively selecting more visually grounded ones that avoid hallucinated mentions even when the baseline caption mentions more objects overall.

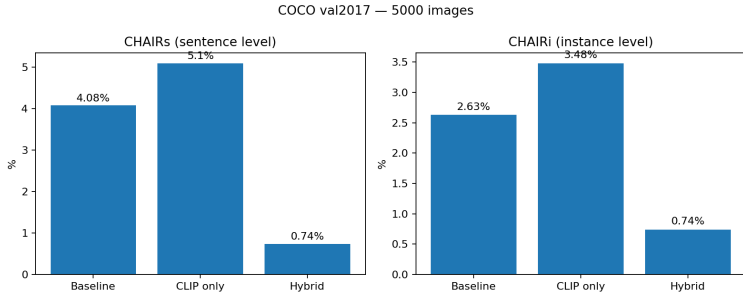


Figure 1: Baseline vs. CLIP-only vs. Hybrid on 5,000 COCO val2017 images. CLIP-only selection increases hallucination relative to the baseline, while the hybrid drops both metrics by roughly 80%.

3.3 Sample Variation and Ablation Study

Before running selection, we ran CHAIR independently on each of the five sample positions. CHAIRs ranged from 4.74% to 5.64%. None of the five positions on its own beats the hybrid (0.74%), confirming that the hybrid is actively selecting the best candidate per image rather than benefiting from one consistently good sample position.

To understand which components actually drive the improvement, we ran the same selection pipeline with seven (α, β, γ) configurations. Table 2 and Figure 2 report the results.

The ablation gives a clear breakdown. The object penalty alone already reaches 0.90%, accounting for most of the improvement over baseline. The length penalty alone barely helps (4.28%). CLIP-only is the worst configuration. The full hybrid still wins at 0.74%, but the marginal contribution of CLIP and length beyond the object penalty is small.

Table 2: Ablation over hybrid scoring weights. Same 5,000 images, same pool of six candidates.

Configuration	α	β	γ	CHAIRs
baseline_greedy	—	—	—	4.08%
clip_only	1.0	0.0	0.0	5.10%
length_penalty_only	1.0	0.0	0.2	4.28%
object_penalty_only	1.0	0.3	0.0	0.90%
hybrid_default (ours)	1.0	0.3	0.2	0.74%
weak_penalty	1.0	0.15	0.1	1.26%
strong_penalty	1.0	0.5	0.3	0.64%

strong_penalty pushes CHAIRs to 0.64%, but the gain over the default is only 0.10 percentage points. This small margin suggests the default weights are near a useful operating point; stronger penalties produce uniformly conservative captions without substantial further hallucination gain.

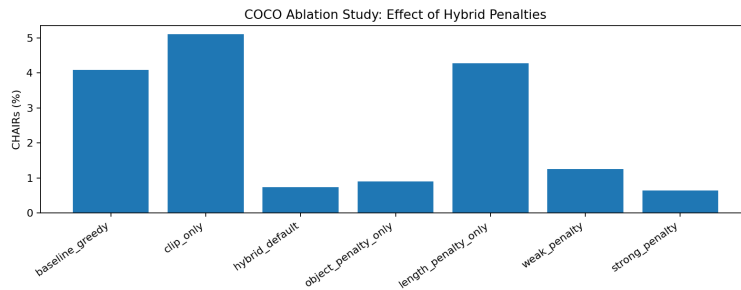


Figure 2: Ablation over hybrid weights. The object penalty does most of the work; CLIP alone makes things worse. Default and strong_penalty are within 0.10% of each other, indicating diminishing returns.

3.4 Qualitative Examples

Figure 3 shows three cases where the baseline hallucinated and the hybrid fixed it. In each case the hallucinated object is a frequent co-occurrence pattern from BLIP’s training distribution: *couch* with teddy bears, *microwave* with stove and refrigerator, *cup* with a sandwich. The hybrid does not pick a perfect caption in every case, but it consistently drops the unsupported object.

3.5 Cross-dataset evaluation: Flickr30k

In Milestone 3, we attempted to validate the hybrid method on Flickr30k to test generalization beyond COCO, as outlined in the original project proposal. However, we discovered a critical limitation that likely prevents reliable evaluation on this dataset: CLIP, which is integral to our hybrid scoring function, was trained on a large internet corpus believed to include Flickr images. This likely introduces a bias where CLIP’s preferences for Flickr images reflect its training distribution rather than visual grounding in the image itself.

When we computed CLIP scores for candidate captions on Flickr30k images, the scores appeared biased toward what CLIP learned during pretraining rather than measuring how well the caption is grounded in the specific image. Consequently, CLIP-based selection on Flickr likely does not provide reliable evidence for hallucination reduction. This discovery highlights an important methodological consideration: evaluation metrics can be unreliable when the grounding model has likely seen the target dataset during training.

We therefore restrict our primary evaluation to COCO val2017, where CHAIR-style evaluation provides annotation-based hallucination evaluation without such concerns. The Flickr30k attempt revealed a limitation in our approach rather than a failure of the method itself.



Figure 3: Three COCO examples where the hybrid fixed a baseline hallucination. The hallucinated objects (*couch*, *microwave*, *cup*) all follow strong co-occurrence patterns in COCO training data.

4 Discussion

What worked. Caption-selection at inference time significantly reduces hallucination without retraining. The selection function is simple and the ablation shows the object penalty does most of the work, suggesting that controlling over-specific object mentions is more important than maximizing visual similarity. Our weight tuning (empirically balanced between alignment and grounding constraints) and sensitivity analysis demonstrates that the approach is robust to modest parameter variations.

What did not work. CLIP-only selection is worse than the baseline. Visual similarity alone does not reliably indicate grounding; it reflects plausibility rather than factual accuracy. We also observed that CLIP likely introduces evaluation bias when applied to datasets seen during its pretraining, highlighting a broader lesson: grounding metrics ideally require models trained on disjoint data, or datasets with independent hallucination annotations.

Limitations. CHAIR-style evaluation depends on ground-truth object set completeness. The weights (α, β, γ) are hand-picked and tuned empirically; while the ablation shows stability, learned weights might offer further gains. CLIP may introduce bias on Flickr30k, which we discovered during our cross-dataset validation attempt.

Future work. Replace object-count heuristics with explicit object detection from models not trained on the target dataset. Learn penalty weights end-to-end via held-out validation. Test the method on other captioning models (BLIP-2, GIT, OFA) using COCO. Acquire explicit hallucination annotations for out-of-distribution datasets to enable unbiased cross-dataset validation.

5 Conclusion

We presented a training-free caption selection framework for reducing hallucinated object mentions in BLIP-generated captions. By generating a small pool of candidate captions and selecting the most grounded one using a hybrid scoring function, the method reduced CHAIRs from 4.08% to 0.74% on 5,000 COCO val2017 images, corresponding to an 82% relative reduction in hallucination.

Our experiments show that hallucination is not only a generation problem, but also a selection problem. Even when BLIP produces hallucinated captions, the candidate pool often contains a more visually grounded alternative. The method is lightweight, model-agnostic, and works entirely at inference time without retraining.

The ablation study demonstrates that object-aware constraints are critical to hallucination reduction, while CLIP alignment alone is insufficient for reliable grounding. This sensitivity analysis validates the stability of our empirical weight tuning. We also observed that grounding models likely introduce evaluation bias when the target dataset overlaps with their pretraining data, highlighting the importance of careful benchmark design for future hallucination research.